MEDIUM · ADVANCED MATH · ANSWER KEY

Nonlinear functions

01

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. $c = 1{,}3001.07^{m}$

Choice B is correct. It's given that in January 2018, there were 1,300 customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a1 + \frac{r}{100}^{m}$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting 1,300 for a and 7 for r in this equation yields $c = 1{,}3001 + \frac{7}{100}^{m}$, or $c = 1{,}3001.07^{m}$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

Q2

15000

The correct answer is $15{,}000$. Let y represent the population of Smithville x years after 1659. Since the population doubled every 75 years, the relationship between x and y can be represented by an exponential equation in the form $y = ab^{\frac{x}{k}}$, where a is the population of Smithville in 1659 and the population increases by a factor of b every k years. It’s given that the population doubled, or increased by a factor of 2, every 75 years. Therefore, $b = 2$ and $k = 75$. Substituting 2 for b and 75 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = a2^{\frac{x}{75}}$. It's also given that the population of this town was $240{,}000$ in 1959. The number of years from 1659 to 1959 is $1959 - 1659$, or 300 years. Therefore, when $x = 300$, the value of y is $240{,}000$. Substituting 300 for x and $240{,}000$ for y in the equation $y = a2^{\frac{x}{75}}$ yields $240{,}000 = a2^{\frac{300}{75}}$, or $240{,}000 = a2^4$, which is equivalent to $240{,}000 = a(16)$. Dividing both sides of this equation by 16 yields $15{,}000 = a$. Therefore, the population of this town in 1659 was $15{,}000$.

Q3

A

A. 2

Choice A is correct. It's given that $pn = 7n^3$. Substituting 56 for pn in this equation yields $56 = 7n^3$. Dividing each side of this equation by 7 yields $8 = n^3$. Taking the cube root of each side of this equation yields $2 = n$. Therefore, when pn is equal to 56, the value of n is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4 C

C. 10

Choice C is correct. It's given that the function f is defined by $fx = x - 4x$. It's also given that $f5 - fa = -15$. Substituting 5 for x in the function $fx = x - 4x$ yields $f5 = 5 - 45$ and substituting a for x in the function $fx = x - 4x$ yields $fa = a - 4a$. Therefore, $f5 = 15$ and $fa = -3a$. Substituting 15 for $f5$ and $-3a$ for fa in the equation $f5 - fa = -15$ yields $15 - -3a = -15$. Subtracting 15 from both sides of this equation yields $--3a = -30$. Dividing both sides of this equation by -1 yields $-3a = 30$. By the definition of absolute value, if $-3a = 30$, then $-3a = 30$ or $-3a = -30$. Dividing both sides of each of these equations by -3 yields $a = -10$ or $a = 10$, respectively. Thus, of the given choices, a value of a that satisfies $f5 - fa = -15$ is 10.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5 3600

The correct answer is 3,600. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting 0 for x in the given equation yields $y = 3{,}600(a)^0$. Since a is positive, this equation is equivalent to $y = 3{,}600(1)$. Therefore, when $x = 0$, the value of y is 3,600. Thus, the predicted number of bacteria initially measured in the growth medium is 3,600.

Q6 2

The correct answer is 2.5. After hitting the ground once, the ball bounces to $20.0 \div 2 = 10.0$ feet. After hitting the ground a second time, the ball bounces to $10.0 \div 2 = 5.0$ feet. After hitting the ground for the third time, the ball bounces to $5.0 \div 2 = 2.5$ feet. Note that 2.5 and 5/2 are examples of ways to enter a correct answer.

Q7

A

A. Approximately 23 grams of the sample remains 87 years after the sample starts to decay.

Choice A is correct. In the given function, st represents the approximate mass, in grams, of the sample that remains t years after the sample starts to decay. It follows that the best interpretation of $s87 = 23$ is that approximately 23 grams of the sample remains 87 years after the sample starts to decay.

Choice B is incorrect. The mass of the sample has decreased by approximately $184 - 23$, or 161, grams, not 23 grams, 87 years after the sample starts to decay.

Choice C is incorrect. The mass of the sample has decreased by approximately 78 grams, not 87 grams, 23 years after the sample starts to decay.

Choice D is incorrect. This would be the best interpretation of $s23 = 87$, not $s87 = 23$.

Q8

A

A. From $x = 0$ to $x = 2$

Choice A is correct. It's given that the variable y represents the height, in meters, of the object above the ground. The graph shows that the height of the object was increasing from $x = 0$ to $x = 2$, and decreasing from $x = 2$ to $x = 4$. Therefore, the height of the object was increasing for the entire interval of time from $x = 0$ to $x = 2$.

Choice B is incorrect. The height of the object wasn't increasing for this entire interval of time, as it was decreasing from $x = 2$ to $x = 4$.

Choice C is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Choice D is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Q9

A

A. The estimated initial beetle population for this species and area in 2012

Choice A is correct. For an exponential function in the form $f x = ab^x$, where a and b are positive constants and $b < 1$, the initial value of $f(x)$, or the value of $f(x)$ when $x = 0$, is a and the value of $f(x)$ decreases by $100(1 - b)\%$ each time x increases by 1. Therefore, the initial value of the function $f x = 4{,}0000.75^x$, or the value of $f(x)$ when $x = 0$, is 4,000. Therefore, the best interpretation of 4,000 in this context is the estimated initial beetle population for this species and area in 2012.

Choice B is incorrect. The estimated beetle population for this species and area 7 years after 2012 is $4{,}000(0.75)^7$, or approximately 534, not 4,000.

Choice C is incorrect. The estimated percent decrease in the beetle population for this species and area each year after 2012 is $100(1 - 0.75)$, or 25, not 4,000.

Choice D is incorrect. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012 is $100\left(1 - 0.75^7\right)$, or approximately 87, not 4,000.

Q10 B

B. 14

Choice B is correct. Let x be the first integer and let y be the second integer. If the first integer is 11 greater than twice the second integer, then $x = 2y + 11$. If the product of the two integers is 546, then $xy = 546$. Substituting $2y + 11$ for x in this equation results in $2y + 11y = 546$. Distributing the y to both terms in the parentheses results in $2y^2 + 11y = 546$. Subtracting 546 from both sides of this equation results in $2y^2 + 11y - 546 = 0$. The left-hand side of this equation can be factored by finding two values whose product is 2-546, or -1,092, and whose sum is 11. The two values whose product is -1,092 and whose sum is 11 are 39 and -28. Thus, the equation $2y^2 + 11y - 546 = 0$ can be rewritten as $2y^2 + 28y - 39y - 546 = 0$, which is equivalent to $2yy - 14 + 39y - 14 = 0$, or $2y + 39y - 14 = 0$. By the zero product property, it follows that $2y + 39 = 0$ and $y - 14 = 0$. Subtracting 39 from both sides of the equation $2y + 39 = 0$ yields $2y = -39$. Dividing both sides of this equation by 2 yields $y = -\frac{39}{2}$. Since y is a positive integer, the value of y is not $-\frac{39}{2}$. Adding 14 to both sides of the equation $y - 14 = 0$ yields $y = 14$. Substituting 14 for y in the equation $xy = 546$ yields $x14 = 546$. Dividing both sides of this equation by 14 results in $x = 39$. Therefore, the two integers are 14 and 39, so the smaller of the two integers is 14.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the larger of the two integers.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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